

# Primary Chain Consolidated Summary

## INTERPRETATION GUIDE

- ADF  $p > 0.05$  and KPSS  $p < 0.05$  -> likely I(1)-like; prefer differences/cointegration models.
- ADF  $p < 0.05$  and KPSS  $p > 0.05$  -> stationary; level models are more admissible.
- Ljung-Box  $p < 0.05$  -> autocorrelation; add lag structure.
- BP/White  $p < 0.05$  -> heteroskedasticity; use robust or HAC standard errors.
- JB  $p < 0.05$  -> non-normality; use robust inference and avoid strict normality assumptions.
- Stability flag=1 -> potential breaks/drift; use rolling or split-sample robustness.
- For retail transmission, compare no-promo vs promo-controlled estimates.

## RUN NOTES

Primary chain is strictly ProducerUA -> ProZorro -> Retail. Retail is estimated separately for silpo, novus, and silpo\_novus. Combined rule: daily median of available silpo and novus standardized\_type prices.

model\_rows=102

pretest\_rows=144

diagnostic\_rows=102

eligibility\_rows=48

Table: Consolidated\_ModelCoefficients

| standardized_type | retailer    | promo_variant    | frequency | link                 | model_family | y_series | x_series | n_obs | sr_coef               | lr_coef              | ect_coef            | ect_pvalue             | asymmetry_short_p   | asymmetry_long_p    | vecm_rank | model_status | unreliable_flag | notes                                                     | primary_chain_doc |
|-------------------|-------------|------------------|-----------|----------------------|--------------|----------|----------|-------|-----------------------|----------------------|---------------------|------------------------|---------------------|---------------------|-----------|--------------|-----------------|-----------------------------------------------------------|-------------------|
| butter            | silpo       | observed         | daily     | producer_to_prozorro | NARDL        | prozorro | producer | 218   | -0.21277642294253607  | 0.27405320585301884  | -0.8083252331134957 | 1.2523745817378955e-26 | 0.787381486847121   | 0.20112635504345666 | nan       | ok           | 0               | Wald short/long asymmetry... Primary chain is strictly... |                   |
| butter            | silpo       | observed         | daily     | prozorro_to_retail   | ARDL         | retail   | prozorro | 41    | 0.2001943602714764    | -0.4257756633486405  | nan                 | nan                    | nan                 | nan                 | nan       | unreliable   | 1               | bounds_lower_p=0.1096; bo... Primary chain is strictly... |                   |
| butter            | silpo       | observed         | daily     | prozorro_to_retail   | NARDL        | retail   | prozorro | 39    | 0.04860976737983126   | -0.12648415712694783 | -0.8756967529857956 | 2.214481682557665e-08  | 0.5941357916590037  | 0.5501601441288803  | nan       | unreliable   | 1               | Wald short/long asymmetry... Primary chain is strictly... |                   |
| butter            | silpo       | promo_controlled | daily     | producer_to_prozorro | NARDL        | prozorro | producer | 218   | -0.21277642294253607  | 0.27405320585301884  | -0.8083252331134957 | 1.2523745817378955e-26 | 0.787381486847121   | 0.20112635504345666 | nan       | ok           | 0               | Wald short/long asymmetry... Primary chain is strictly... |                   |
| butter            | silpo       | promo_controlled | daily     | prozorro_to_retail   | ARDL         | retail   | prozorro | 41    | 0.2001943602714764    | -0.4257756633486405  | nan                 | nan                    | nan                 | nan                 | nan       | unreliable   | 1               | bounds_lower_p=0.1096; bo... Primary chain is strictly... |                   |
| butter            | silpo       | promo_controlled | daily     | prozorro_to_retail   | NARDL        | retail   | prozorro | 39    | 0.04860976737983126   | -0.12648415712694783 | -0.8756967529857956 | 2.214481682557665e-08  | 0.5941357916590037  | 0.5501601441288803  | nan       | unreliable   | 1               | Wald short/long asymmetry... Primary chain is strictly... |                   |
| butter            | novus       | observed         | daily     | producer_to_prozorro | NARDL        | prozorro | producer | 218   | -0.21277642294253607  | 0.27405320585301884  | -0.8083252331134957 | 1.2523745817378955e-26 | 0.787381486847121   | 0.20112635504345666 | nan       | ok           | 0               | Wald short/long asymmetry... Primary chain is strictly... |                   |
| butter            | novus       | promo_controlled | daily     | producer_to_prozorro | NARDL        | prozorro | producer | 218   | -0.21277642294253607  | 0.27405320585301884  | -0.8083252331134957 | 1.2523745817378955e-26 | 0.787381486847121   | 0.20112635504345666 | nan       | ok           | 0               | Wald short/long asymmetry... Primary chain is strictly... |                   |
| butter            | silpo_novus | observed         | daily     | producer_to_prozorro | NARDL        | prozorro | producer | 218   | -0.21277642294253607  | 0.27405320585301884  | -0.8083252331134957 | 1.2523745817378955e-26 | 0.787381486847121   | 0.20112635504345666 | nan       | ok           | 0               | Wald short/long asymmetry... Primary chain is strictly... |                   |
| butter            | silpo_novus | observed         | daily     | prozorro_to_retail   | ARDL         | retail   | prozorro | 51    | 0.3808399108271916    | 5.038120798935976    | nan                 | nan                    | nan                 | nan                 | nan       | ok           | 0               | bounds_lower_p=0.02219; b... Primary chain is strictly... |                   |
| butter            | silpo_novus | observed         | daily     | prozorro_to_retail   | ECM          | retail   | prozorro | 49    | -0.019317533092633468 | -0.11304424771502886 | -0.5842642909954745 | 0.007904731650635205   | nan                 | nan                 | nan       | ok           | 0               | engle_granger_p=0.06; lag... Primary chain is strictly... |                   |
| butter            | silpo_novus | observed         | daily     | prozorro_to_retail   | NARDL        | retail   | prozorro | 49    | 0.03764617386887498   | -0.8051638051994185  | -0.934681111534598  | 0.0012434236716784065  | 0.9262473301905363  | 0.3206536900888561  | nan       | ok           | 0               | Wald short/long asymmetry... Primary chain is strictly... |                   |
| butter            | silpo_novus | promo_controlled | daily     | producer_to_prozorro | NARDL        | prozorro | producer | 218   | -0.21277642294253607  | 0.27405320585301884  | -0.8083252331134957 | 1.2523745817378955e-26 | 0.787381486847121   | 0.20112635504345666 | nan       | ok           | 0               | Wald short/long asymmetry... Primary chain is strictly... |                   |
| butter            | silpo_novus | promo_controlled | daily     | prozorro_to_retail   | ARDL         | retail   | prozorro | 51    | 0.3808399108271916    | 5.038120798935976    | nan                 | nan                    | nan                 | nan                 | nan       | ok           | 0               | bounds_lower_p=0.02219; b... Primary chain is strictly... |                   |
| butter            | silpo_novus | promo_controlled | daily     | prozorro_to_retail   | ECM          | retail   | prozorro | 49    | -0.019317533092633468 | -0.11304424771502886 | -0.5842642909954745 | 0.007904731650635205   | nan                 | nan                 | nan       | ok           | 0               | engle_granger_p=0.06; lag... Primary chain is strictly... |                   |
| butter            | silpo_novus | promo_controlled | daily     | prozorro_to_retail   | NARDL        | retail   | prozorro | 49    | 0.03764617386887498   | -0.8051638051994185  | -0.934681111534598  | 0.0012434236716784065  | 0.9262473301905363  | 0.3206536900888561  | nan       | ok           | 0               | Wald short/long asymmetry... Primary chain is strictly... |                   |
| cottage_cheese    | silpo       | observed         | daily     | none                 | none         | n/a      | n/a      | 203   | nan                   | nan                  | nan                 | nan                    | nan                 | nan                 | nan       | unavailable  | 1               | No admissible model estim... Primary chain is strictly... |                   |
| cottage_cheese    | silpo       | promo_controlled | daily     | none                 | none         | n/a      | n/a      | 203   | nan                   | nan                  | nan                 | nan                    | nan                 | nan                 | nan       | unavailable  | 1               | No admissible model estim... Primary chain is strictly... |                   |
| cottage_cheese    | novus       | observed         | daily     | none                 | none         | n/a      | n/a      | 203   | nan                   | nan                  | nan                 | nan                    | nan                 | nan                 | nan       | unavailable  | 1               | No admissible model estim... Primary chain is strictly... |                   |
| cottage_cheese    | novus       | promo_controlled | daily     | none                 | none         | n/a      | n/a      | 203   | nan                   | nan                  | nan                 | nan                    | nan                 | nan                 | nan       | unavailable  | 1               | No admissible model estim... Primary chain is strictly... |                   |
| cottage_cheese    | silpo_novus | observed         | daily     | prozorro_to_retail   | ARDL         | retail   | prozorro | 42    | -0.4066263125016476   | -1.685428485355531   | nan                 | nan                    | nan                 | nan                 | nan       | ok           | 0               | bounds_lower_p=0.02475; b... Primary chain is strictly... |                   |
| cottage_cheese    | silpo_novus | observed         | daily     | prozorro_to_retail   | NARDL        | retail   | prozorro | 40    | -0.04779905990700553  | -0.5494671968185126  | -0.9346201147084174 | 1.5479768354950692e-10 | 0.8868176751443891  | 0.32025906973636026 | nan       | ok           | 0               | Wald short/long asymmetry... Primary chain is strictly... |                   |
| cottage_cheese    | silpo_novus | promo_controlled | daily     | prozorro_to_retail   | ARDL         | retail   | prozorro | 42    | -0.4066263125016476   | -1.685428485355531   | nan                 | nan                    | nan                 | nan                 | nan       | ok           | 0               | bounds_lower_p=0.02475; b... Primary chain is strictly... |                   |
| cottage_cheese    | silpo_novus | promo_controlled | daily     | prozorro_to_retail   | NARDL        | retail   | prozorro | 40    | -0.04779905990700553  | -0.5494671968185126  | -0.9346201147084174 | 1.5479768354950692e-10 | 0.8868176751443891  | 0.32025906973636026 | nan       | ok           | 0               | Wald short/long asymmetry... Primary chain is strictly... |                   |
| cream             | silpo       | observed         | daily     | producer_to_prozorro | NARDL        | prozorro | producer | 197   | 3.7936802523963236    | 0.33453408711897387  | -1.002572700947699  | 2.7759014458990185e-16 | 0.07891071636138866 | 0.31445927128489587 | nan       | ok           | 0               | Wald short/long asymmetry... Primary chain is strictly... |                   |
| cream             | silpo       | observed         | daily     | prozorro_to_retail   | ARDL         | retail   | prozorro | 42    | -0.027190870499124387 | -0.2713968465337292  | nan                 | nan                    | nan                 | nan                 | nan       | unreliable   | 1               | bounds_lower_p=0.09399; b... Primary chain is strictly... |                   |
| cream             | silpo       | observed         | daily     | prozorro_to_retail   | NARDL        | retail   | prozorro | 40    | -0.070469572016711    | -0.23542918833682522 | -0.539409399232454  | 0.0003540689684816318  | 0.23639756571220905 | 0.05201532870115675 | nan       | ok           | 0               | Wald short/long asymmetry... Primary chain is strictly... |                   |
| cream             | silpo       | promo_controlled | daily     | producer_to_prozorro | NARDL        | prozorro | producer | 197   | 3.7936802523963236    | 0.33453408711897387  | -1.002572700947699  | 2.7759014458990185e-16 | 0.07891071636138866 | 0.31445927128489587 | nan       | ok           | 0               | Wald short/long asymmetry... Primary chain is strictly... |                   |
| cream             | silpo       | promo_controlled | daily     | prozorro_to_retail   | ARDL         | retail   | prozorro | 42    | -0.027190870499124387 | -0.2713968465337292  | nan                 | nan                    | nan                 | nan                 | nan       | unreliable   | 1               | bounds_lower_p=0.09399; b... Primary chain is strictly... |                   |
| cream             | silpo       | promo_controlled | daily     | prozorro_to_retail   | NARDL        | retail   | prozorro | 40    | -0.070469572016711    | -0.23542918833682522 | -0.539409399232454  | 0.0003540689684816318  | 0.23639756571220905 | 0.05201532870115675 | nan       | ok           | 0               | Wald short/long asymmetry... Primary chain is strictly... |                   |
| cream             | novus       | observed         | daily     | producer_to_prozorro | NARDL        | prozorro | producer | 197   | 3.7936802523963236    | 0.33453408711897387  | -1.002572700947699  | 2.7759014458990185e-16 | 0.07891071636138866 | 0.31445927128489587 | nan       | ok           | 0               | Wald short/long asymmetry... Primary chain is strictly... |                   |
| cream             | novus       | promo_controlled | daily     | producer_to_prozorro | NARDL        | prozorro | producer | 197   | 3.7936802523963236    | 0.33453408711897387  | -1.002572700947699  | 2.7759014458990185e-16 | 0.07891071636138866 | 0.31445927128489587 | nan       | ok           | 0               | Wald short/long asymmetry... Primary chain is strictly... |                   |
| cream             | silpo_novus | observed         | daily     | none                 | none         | n/a      | n/a      | 1803  | nan                   | nan                  | nan                 | nan                    | nan                 | nan                 | nan       | unavailable  | 1               | No admissible model estim... Primary chain is strictly... |                   |
| cream             | silpo_novus | promo_controlled | daily     | none                 | none         | n/a      | n/a      | 1803  | nan                   | nan                  | nan                 | nan                    | nan                 | nan                 | nan       | unavailable  | 1               | No admissible model estim... Primary chain is strictly... |                   |
| hard_cheese       | silpo       | observed         | daily     | producer_to_prozorro | NARDL        | prozorro | producer | 187   | -1.901684280673745    | 0.31953965595429906  | -0.703887241265392  | 2.747609431022365e-15  | 0.16253431119720008 | 0.2610022514748195  | nan       | ok           | 0               | Wald short/long asymmetry... Primary chain is strictly... |                   |
| hard_cheese       | silpo       | observed         | daily     | prozorro_to_retail   | ARDL         | retail   | prozorro | 38    | 1.0453508797868807    | 16.32664882022445    | nan                 | nan                    | nan                 | nan                 | nan       | ok           | 0               | bounds_lower_p=nan; bound... Primary chain is strictly... |                   |
| hard_cheese       | silpo       | observed         | daily     | prozorro_to_retail   | NARDL        | retail   | prozorro | 36    | 0.030935387091438615  | 0.01877747610695744  | -1.1313012234118889 | 2.5442359295873624e-08 | 0.8109745337593639  | 0.9240474765408485  | nan       | ok           | 0               | Wald short/long asymmetry... Primary chain is strictly... |                   |
| hard_cheese       | silpo       | promo_controlled | daily     | producer_to_prozorro | NARDL        | prozorro | producer | 187   | -1.901684280673745    | 0.31953965595429906  | -0.703887241265392  | 2.747609431022365e-15  | 0.16253431119720008 | 0.2610022514748195  | nan       | ok           | 0               | Wald short/long asymmetry... Primary chain is strictly... |                   |
| hard_cheese       | silpo       | promo_controlled | daily     | prozorro_to_retail   | ARDL         | retail   | prozorro | 38    | 1.0453508797868807    | 16.32664882022445    | nan                 | nan                    | nan                 | nan                 | nan       | ok           | 0               | bounds_lower_p=nan; bound... Primary chain is strictly... |                   |
| hard_cheese       | silpo       | promo_controlled | daily     | prozorro_to_retail   | NARDL        | retail   | prozorro | 36    | 0.030935387091438615  | 0.01877747610695744  | -1.1313012234118889 | 2.5442359295873624e-08 | 0.8109745337593639  | 0.9240474765408485  | nan       | ok           | 0               | Wald short/long asymmetry... Primary chain is strictly... |                   |

Table: Consolidated\_PreTests

| series   | adf_level_p            | kpss_level_p         | adf_diff1_p            | kpss_diff1_p       | adf_diff2_p            | kpss_diff2_p          | integration_class | stability_flag | standardized_type | retailer    | promo_variant    |
|----------|------------------------|----------------------|------------------------|--------------------|------------------------|-----------------------|-------------------|----------------|-------------------|-------------|------------------|
| producer | 0.08225840900715364    | 0.01                 | 2.681802621264837e-12  | 0.1                | 1.3006957963952123e-20 | 0.1                   | I(1)              | 1              | butter            | silpo       | observed         |
| prozorro | 0.056404226942647726   | 0.01                 | 3.518946761497874e-22  | 0.1                | 1.9269184182026386e-13 | 0.1                   | I(1)              | 1              | butter            | silpo       | observed         |
| retail   | 2.2989340931225115e-06 | 0.1                  | 2.5595238291844733e-09 | 0.1                | 8.70860721185971e-05   | 0.041666666666666662  | I(0)              | 0              | butter            | silpo       | observed         |
| producer | 0.08225840900715364    | 0.01                 | 2.681802621264837e-12  | 0.1                | 1.3006957963952123e-20 | 0.1                   | I(1)              | 1              | butter            | silpo       | promo_controlled |
| prozorro | 0.056404226942647726   | 0.01                 | 3.518946761497874e-22  | 0.1                | 1.9269184182026386e-13 | 0.1                   | I(1)              | 1              | butter            | silpo       | promo_controlled |
| retail   | 2.2989340931225115e-06 | 0.1                  | 2.5595238291844733e-09 | 0.1                | 8.70860721185971e-05   | 0.041666666666666662  | I(0)              | 0              | butter            | silpo       | promo_controlled |
| producer | 0.08225840900715364    | 0.01                 | 2.681802621264837e-12  | 0.1                | 1.3006957963952123e-20 | 0.1                   | I(1)              | 1              | butter            | novus       | observed         |
| prozorro | 0.056404226942647726   | 0.01                 | 3.518946761497874e-22  | 0.1                | 1.9269184182026386e-13 | 0.1                   | I(1)              | 1              | butter            | novus       | observed         |
| retail   | nan                    | nan                  | nan                    | nan                | nan                    | nan                   | ambiguous         | 0              | butter            | novus       | observed         |
| producer | 0.08225840900715364    | 0.01                 | 2.681802621264837e-12  | 0.1                | 1.3006957963952123e-20 | 0.1                   | I(1)              | 1              | butter            | novus       | promo_controlled |
| prozorro | 0.056404226942647726   | 0.01                 | 3.518946761497874e-22  | 0.1                | 1.9269184182026386e-13 | 0.1                   | I(1)              | 1              | butter            | novus       | promo_controlled |
| retail   | nan                    | nan                  | nan                    | nan                | nan                    | nan                   | ambiguous         | 0              | butter            | novus       | promo_controlled |
| producer | 0.08225840900715364    | 0.01                 | 2.681802621264837e-12  | 0.1                | 1.3006957963952123e-20 | 0.1                   | I(1)              | 1              | butter            | silpo_novus | observed         |
| prozorro | 0.056404226942647726   | 0.01                 | 3.518946761497874e-22  | 0.1                | 1.9269184182026386e-13 | 0.1                   | I(1)              | 1              | butter            | silpo_novus | observed         |
| retail   | 0.009658462541496323   | 0.015275724253247233 | 0.0004241747553426809  | 0.1                | 0.0016100992365552236  | 0.1                   | I(1)              | 0              | butter            | silpo_novus | observed         |
| producer | 0.08225840900715364    | 0.01                 | 2.681802621264837e-12  | 0.1                | 1.3006957963952123e-20 | 0.1                   | I(1)              | 1              | butter            | silpo_novus | promo_controlled |
| prozorro | 0.056404226942647726   | 0.01                 | 3.518946761497874e-22  | 0.1                | 1.9269184182026386e-13 | 0.1                   | I(1)              | 1              | butter            | silpo_novus | promo_controlled |
| retail   | 0.009658462541496323   | 0.015275724253247233 | 0.0004241747553426809  | 0.1                | 0.0016100992365552236  | 0.1                   | I(1)              | 0              | butter            | silpo_novus | promo_controlled |
| producer | nan                    | nan                  | nan                    | nan                | nan                    | nan                   | ambiguous         | 0              | cottage_cheese    | silpo       | observed         |
| prozorro | 1.1483227108295172e-10 | 0.01                 | 4.589982408975803e-17  | 0.1                | 4.813752592554151e-10  | 0.1                   | I(1)              | 0              | cottage_cheese    | silpo       | observed         |
| retail   | 5.9157706543378735e-08 | 0.1                  | 7.517018918246667e-13  | 0.0865089553841217 | 0.001856917684812617   | 0.0416666666666666796 | I(0)              | 0              | cottage_cheese    | silpo       | observed         |
| producer | nan                    | nan                  | nan                    | nan                | nan                    | nan                   | ambiguous         | 0              | cottage_cheese    | silpo       | promo_controlled |
| prozorro | 1.1483227108295172e-10 | 0.01                 | 4.589982408975803e-17  | 0.1                | 4.813752592554151e-10  | 0.1                   | I(1)              | 0              | cottage_cheese    | silpo       | promo_controlled |
| retail   | 5.9157706543378735e-08 | 0.1                  | 7.517018918246667e-13  | 0.0865089553841217 | 0.001856917684812617   | 0.0416666666666666796 | I(0)              | 0              | cottage_cheese    | silpo       | promo_controlled |
| producer | nan                    | nan                  | nan                    | nan                | nan                    | nan                   | ambiguous         | 0              | cottage_cheese    | novus       | observed         |
| prozorro | 1.1483227108295172e-10 | 0.01                 | 4.589982408975803e-17  | 0.1                | 4.813752592554151e-10  | 0.1                   | I(1)              | 0              | cottage_cheese    | novus       | observed         |
| retail   | nan                    | nan                  | nan                    | nan                | nan                    | nan                   | ambiguous         | 0              | cottage_cheese    | novus       | observed         |
| producer | nan                    | nan                  | nan                    | nan                | nan                    | nan                   | ambiguous         | 0              | cottage_cheese    | novus       | promo_controlled |
| prozorro | 1.1483227108295172e-10 | 0.01                 | 4.589982408975803e-17  | 0.1                | 4.813752592554151e-10  | 0.1                   | I(1)              | 0              | cottage_cheese    | novus       | promo_controlled |
| retail   | nan                    | nan                  | nan                    | nan                | nan                    | nan                   | ambiguous         | 0              | cottage_cheese    | novus       | promo_controlled |
| producer | nan                    | nan                  | nan                    | nan                | nan                    | nan                   | ambiguous         | 0              | cottage_cheese    | silpo_novus | observed         |
| prozorro | 1.1483227108295172e-10 | 0.01                 | 4.589982408975803e-17  | 0.1                | 4.813752592554151e-10  | 0.1                   | I(1)              | 0              | cottage_cheese    | silpo_novus | observed         |
| retail   | 7.053635234015645e-08  | 0.055181023671470035 | 0.0008960877939895775  | 0.1                | 4.783037514608705e-06  | 0.1                   | I(0)              | 0              | cottage_cheese    | silpo_novus | observed         |
| producer | nan                    | nan                  | nan                    | nan                | nan                    | nan                   | ambiguous         | 0              | cottage_cheese    | silpo_novus | promo_controlled |
| prozorro | 1.1483227108295172e-10 | 0.01                 | 4.589982408975803e-17  | 0.1                | 4.813752592554151e-10  | 0.1                   | I(1)              | 0              | cottage_cheese    | silpo_novus | promo_controlled |
| retail   | 7.053635234015645e-08  | 0.055181023671470035 | 0.0008960877939895775  | 0.1                | 4.783037514608705e-06  | 0.1                   | I(0)              | 0              | cottage_cheese    | silpo_novus | promo_controlled |
| producer | 3.1663521310609106e-05 | 0.01                 | 1.807358615624225e-13  | 0.1                | 2.250471045105603e-20  | 0.1                   | I(1)              | 1              | cream             | silpo       | observed         |
| prozorro | 5.15934840442518e-22   | 0.1                  | 3.6046356335359884e-13 | 0.1                | 1.1693001750830719e-11 | 0.09309066682789993   | I(0)              | 0              | cream             | silpo       | observed         |
| retail   | 0.0801729662500495     | 0.1                  | 0.0004821332512077595  | 0.1                | 4.135571752859827e-05  | 0.1                   | I(1)              | 0              | cream             | silpo       | observed         |
| producer | 3.1663521310609106e-05 | 0.01                 | 1.807358615624225e-13  | 0.1                | 2.250471045105603e-20  | 0.1                   | I(1)              | 1              | cream             | silpo       | promo_controlled |

Table: Consolidated\_ResidualDiagnostics

| model_family | link                 | y_series | x_series | ljungbox_p           | arch_p               | jb_p                   | unreliable_flag | standardized_type | retailer    | promo_variant    |
|--------------|----------------------|----------|----------|----------------------|----------------------|------------------------|-----------------|-------------------|-------------|------------------|
| NARDL        | producer_to_prozorro | prozorro | producer | 0.3043740439991276   | 0.5287391650637272   | 4.2203813853019655e-84 | 0               | butter            | silpo       | observed         |
| ARDL         | prozorro_to_retail   | retail   | prozorro | 0.01144894530681835  | 0.09344174279103216  | 0.6443904339549587     | 1               | butter            | silpo       | observed         |
| NARDL        | prozorro_to_retail   | retail   | prozorro | 0.5591712653384464   | 0.046665474894684277 | 0.8050949849331446     | 1               | butter            | silpo       | observed         |
| NARDL        | producer_to_prozorro | prozorro | producer | 0.3043740439991276   | 0.5287391650637272   | 4.2203813853019655e-84 | 0               | butter            | silpo       | promo_controlled |
| ARDL         | prozorro_to_retail   | retail   | prozorro | 0.01144894530681835  | 0.09344174279103216  | 0.6443904339549587     | 1               | butter            | silpo       | promo_controlled |
| NARDL        | prozorro_to_retail   | retail   | prozorro | 0.5591712653384464   | 0.046665474894684277 | 0.8050949849331446     | 1               | butter            | silpo       | promo_controlled |
| NARDL        | producer_to_prozorro | prozorro | producer | 0.3043740439991276   | 0.5287391650637272   | 4.2203813853019655e-84 | 0               | butter            | novus       | observed         |
| NARDL        | producer_to_prozorro | prozorro | producer | 0.3043740439991276   | 0.5287391650637272   | 4.2203813853019655e-84 | 0               | butter            | novus       | promo_controlled |
| NARDL        | producer_to_prozorro | prozorro | producer | 0.3043740439991276   | 0.5287391650637272   | 4.2203813853019655e-84 | 0               | butter            | silpo_novus | observed         |
| ARDL         | prozorro_to_retail   | retail   | prozorro | 0.1517137359967815   | 0.9428759814198608   | 0.12163941337757182    | 0               | butter            | silpo_novus | observed         |
| ECM          | prozorro_to_retail   | retail   | prozorro | 0.7561861082620147   | 0.1308148001577336   | 1.4411688625167337e-05 | 0               | butter            | silpo_novus | observed         |
| NARDL        | prozorro_to_retail   | retail   | prozorro | 0.4526018945159763   | 0.17734130295808273  | 0.0008392768825094401  | 0               | butter            | silpo_novus | observed         |
| NARDL        | producer_to_prozorro | prozorro | producer | 0.3043740439991276   | 0.5287391650637272   | 4.2203813853019655e-84 | 0               | butter            | silpo_novus | promo_controlled |
| ARDL         | prozorro_to_retail   | retail   | prozorro | 0.1517137359967815   | 0.9428759814198608   | 0.12163941337757182    | 0               | butter            | silpo_novus | promo_controlled |
| ECM          | prozorro_to_retail   | retail   | prozorro | 0.7561861082620147   | 0.1308148001577336   | 1.4411688625167337e-05 | 0               | butter            | silpo_novus | promo_controlled |
| NARDL        | prozorro_to_retail   | retail   | prozorro | 0.4526018945159763   | 0.17734130295808273  | 0.0008392768825094401  | 0               | butter            | silpo_novus | promo_controlled |
| none         | none                 | n/a      | n/a      | nan                  | nan                  | nan                    | 1               | cottage_cheese    | silpo       | observed         |
| none         | none                 | n/a      | n/a      | nan                  | nan                  | nan                    | 1               | cottage_cheese    | silpo       | promo_controlled |
| none         | none                 | n/a      | n/a      | nan                  | nan                  | nan                    | 1               | cottage_cheese    | novus       | observed         |
| none         | none                 | n/a      | n/a      | nan                  | nan                  | nan                    | 1               | cottage_cheese    | novus       | promo_controlled |
| ARDL         | prozorro_to_retail   | retail   | prozorro | 0.1007447696043067   | 0.45591024880877795  | 0.7986167356499732     | 0               | cottage_cheese    | silpo_novus | observed         |
| NARDL        | prozorro_to_retail   | retail   | prozorro | 0.13802457343734706  | 0.7478438770300309   | 2.6950380978724114e-06 | 0               | cottage_cheese    | silpo_novus | observed         |
| ARDL         | prozorro_to_retail   | retail   | prozorro | 0.1007447696043067   | 0.45591024880877795  | 0.7986167356499732     | 0               | cottage_cheese    | silpo_novus | promo_controlled |
| NARDL        | prozorro_to_retail   | retail   | prozorro | 0.13802457343734706  | 0.7478438770300309   | 2.6950380978724114e-06 | 0               | cottage_cheese    | silpo_novus | promo_controlled |
| NARDL        | producer_to_prozorro | prozorro | producer | 0.889712370853929    | 0.2018078467373643   | 1.393192716168525e-262 | 0               | cream             | silpo       | observed         |
| ARDL         | prozorro_to_retail   | retail   | prozorro | 0.004919229359110526 | 0.5840482434798735   | 0.8870969508504547     | 1               | cream             | silpo       | observed         |
| NARDL        | prozorro_to_retail   | retail   | prozorro | 0.5530832153948084   | 0.6339111636110053   | 0.7966413904775785     | 0               | cream             | silpo       | observed         |
| NARDL        | producer_to_prozorro | prozorro | producer | 0.889712370853929    | 0.2018078467373643   | 1.393192716168525e-262 | 0               | cream             | silpo       | promo_controlled |
| ARDL         | prozorro_to_retail   | retail   | prozorro | 0.004919229359110526 | 0.5840482434798735   | 0.8870969508504547     | 1               | cream             | silpo       | promo_controlled |
| NARDL        | prozorro_to_retail   | retail   | prozorro | 0.5530832153948084   | 0.6339111636110053   | 0.7966413904775785     | 0               | cream             | silpo       | promo_controlled |
| NARDL        | producer_to_prozorro | prozorro | producer | 0.889712370853929    | 0.2018078467373643   | 1.393192716168525e-262 | 0               | cream             | novus       | observed         |
| NARDL        | producer_to_prozorro | prozorro | producer | 0.889712370853929    | 0.2018078467373643   | 1.393192716168525e-262 | 0               | cream             | novus       | promo_controlled |
| none         | none                 | n/a      | n/a      | nan                  | nan                  | nan                    | 1               | cream             | silpo_novus | observed         |
| none         | none                 | n/a      | n/a      | nan                  | nan                  | nan                    | 1               | cream             | silpo_novus | promo_controlled |
| NARDL        | producer_to_prozorro | prozorro | producer | 0.30453926702142564  | 0.2688968463348229   | 0.05424410826903407    | 0               | hard_cheese       | silpo       | observed         |
| ARDL         | prozorro_to_retail   | retail   | prozorro | 0.644080315519358    | 1.0                  | 0.0002471109442311297  | 0               | hard_cheese       | silpo       | observed         |
| NARDL        | prozorro_to_retail   | retail   | prozorro | 0.4999098230155331   | 0.5136560684499863   | 0.22218374478505765    | 0               | hard_cheese       | silpo       | observed         |
| NARDL        | producer_to_prozorro | prozorro | producer | 0.30453926702142564  | 0.2688968463348229   | 0.05424410826903407    | 0               | hard_cheese       | silpo       | promo_controlled |
| ARDL         | prozorro_to_retail   | retail   | prozorro | 0.644080315519358    | 1.0                  | 0.0002471109442311297  | 0               | hard_cheese       | silpo       | promo_controlled |
| NARDL        | prozorro_to_retail   | retail   | prozorro | 0.4999098230155331   | 0.5136560684499863   | 0.22218374478505765    | 0               | hard_cheese       | silpo       | promo_controlled |

Table: Consolidated\_Eligibility

| standardized_type | retailer    | promo_variant    | frequency | integration_producer | integration_prozorro | integration_retail | n_obs_producer | n_obs_prozorro | n_obs_retail | n_obs_pair_prod_prozorro | n_obs_pair_prozorro_retail | any_i2 | eligibility_note             |
|-------------------|-------------|------------------|-----------|----------------------|----------------------|--------------------|----------------|----------------|--------------|--------------------------|----------------------------|--------|------------------------------|
| butter            | silpo       | observed         | daily     | I(1)                 | I(1)                 | I(0)               | 1793           | 230            | 48           | 220                      | 41                         | 0      | Admissible for ARDL/ECM s... |
|                   | silpo       | promo_controlled | daily     | I(1)                 | I(1)                 | I(0)               | 1793           | 230            | 48           | 220                      | 41                         | 0      | Admissible for ARDL/ECM s... |
|                   | novus       | observed         | daily     | I(1)                 | I(1)                 | ambiguous          | 1793           | 230            | 19           | 220                      | 17                         | 0      | Low pair overlap in main ... |
|                   | novus       | promo_controlled | daily     | I(1)                 | I(1)                 | ambiguous          | 1793           | 230            | 19           | 220                      | 17                         | 0      | Low pair overlap in main ... |
|                   | silpo_novus | observed         | daily     | I(1)                 | I(1)                 | I(1)               | 1793           | 230            | 58           | 220                      | 51                         | 0      | Admissible for ARDL/ECM s... |
|                   | silpo_novus | promo_controlled | daily     | I(1)                 | I(1)                 | I(1)               | 1793           | 230            | 58           | 220                      | 51                         | 0      | Admissible for ARDL/ECM s... |
| cottage_cheese    | silpo       | observed         | daily     | ambiguous            | I(1)                 | I(0)               | 0              | 190            | 48           | 0                        | 35                         | 0      | Low pair overlap in main ... |
| cottage_cheese    | silpo       | promo_controlled | daily     | ambiguous            | I(1)                 | I(0)               | 0              | 190            | 48           | 0                        | 35                         | 0      | Low pair overlap in main ... |
| cottage_cheese    | novus       | observed         | daily     | ambiguous            | I(1)                 | ambiguous          | 0              | 190            | 20           | 0                        | 17                         | 0      | Low pair overlap in main ... |
| cottage_cheese    | novus       | promo_controlled | daily     | ambiguous            | I(1)                 | ambiguous          | 0              | 190            | 20           | 0                        | 17                         | 0      | Low pair overlap in main ... |
| cottage_cheese    | silpo_novus | observed         | daily     | ambiguous            | I(1)                 | I(0)               | 0              | 190            | 55           | 0                        | 42                         | 0      | Low pair overlap in main ... |
| cottage_cheese    | silpo_novus | promo_controlled | daily     | ambiguous            | I(1)                 | I(0)               | 0              | 190            | 55           | 0                        | 42                         | 0      | Low pair overlap in main ... |
| cream             | silpo       | observed         | daily     | I(1)                 | I(0)                 | I(1)               | 1793           | 209            | 48           | 199                      | 42                         | 0      | Admissible for ARDL/ECM s... |
|                   | silpo       | promo_controlled | daily     | I(1)                 | I(0)                 | I(1)               | 1793           | 209            | 48           | 199                      | 42                         | 0      | Admissible for ARDL/ECM s... |
|                   | novus       | observed         | daily     | I(1)                 | I(0)                 | ambiguous          | 1793           | 209            | 18           | 199                      | 16                         | 0      | Low pair overlap in main ... |
|                   | novus       | promo_controlled | daily     | I(1)                 | I(0)                 | ambiguous          | 1793           | 209            | 18           | 199                      | 16                         | 0      | Low pair overlap in main ... |
|                   | silpo_novus | observed         | daily     | I(1)                 | I(0)                 | I(2)               | 1793           | 209            | 55           | 199                      | 48                         | 1      | Rejected for ARDL/ECM due... |
|                   | silpo_novus | promo_controlled | daily     | I(1)                 | I(0)                 | I(2)               | 1793           | 209            | 55           | 199                      | 48                         | 1      | Rejected for ARDL/ECM due... |
| hard_cheese       | silpo       | observed         | daily     | I(1)                 | I(0)                 | I(1)               | 1793           | 199            | 48           | 189                      | 38                         | 0      | Admissible for ARDL/ECM s... |
| hard_cheese       | silpo       | promo_controlled | daily     | I(1)                 | I(0)                 | I(1)               | 1793           | 199            | 48           | 189                      | 38                         | 0      | Admissible for ARDL/ECM s... |
| hard_cheese       | novus       | observed         | daily     | I(1)                 | I(0)                 | I(0)               | 1793           | 199            | 37           | 189                      | 32                         | 0      | Admissible for ARDL/ECM s... |
| hard_cheese       | novus       | promo_controlled | daily     | I(1)                 | I(0)                 | I(0)               | 1793           | 199            | 37           | 189                      | 32                         | 0      | Admissible for ARDL/ECM s... |
| hard_cheese       | silpo_novus | observed         | daily     | I(1)                 | I(0)                 | I(0)               | 1793           | 199            | 60           | 189                      | 49                         | 0      | Admissible for ARDL/ECM s... |
| hard_cheese       | silpo_novus | promo_controlled | daily     | I(1)                 | I(0)                 | I(0)               | 1793           | 199            | 60           | 189                      | 49                         | 0      | Admissible for ARDL/ECM s... |
| milk              | silpo       | observed         | daily     | I(1)                 | I(1)                 | I(1)               | 1793           | 211            | 48           | 201                      | 37                         | 0      | Admissible for ARDL/ECM s... |
|                   | silpo       | promo_controlled | daily     | I(1)                 | I(1)                 | I(1)               | 1793           | 211            | 48           | 201                      | 37                         | 0      | Admissible for ARDL/ECM s... |
|                   | novus       | observed         | daily     | I(1)                 | I(1)                 | ambiguous          | 1793           | 211            | 22           | 201                      | 20                         | 0      | Low pair overlap in main ... |
|                   | novus       | promo_controlled | daily     | I(1)                 | I(1)                 | ambiguous          | 1793           | 211            | 22           | 201                      | 20                         | 0      | Low pair overlap in main ... |
|                   | silpo_novus | observed         | daily     | I(1)                 | I(1)                 | I(2)               | 1793           | 211            | 59           | 201                      | 48                         | 1      | Rejected for ARDL/ECM due... |
|                   | silpo_novus | promo_controlled | daily     | I(1)                 | I(1)                 | I(2)               | 1793           | 211            | 59           | 201                      | 48                         | 1      | Rejected for ARDL/ECM due... |
| other             | silpo       | observed         | daily     | ambiguous            | I(1)                 | I(1)               | 0              | 232            | 54           | 0                        | 50                         | 0      | Low pair overlap in main ... |
| other             | silpo       | promo_controlled | daily     | ambiguous            | I(1)                 | I(1)               | 0              | 232            | 54           | 0                        | 50                         | 0      | Low pair overlap in main ... |
| other             | novus       | observed         | daily     | ambiguous            | I(1)                 | ambiguous          | 0              | 232            | 7            | 0                        | 7                          | 0      | Low pair overlap in main ... |
| other             | novus       | promo_controlled | daily     | ambiguous            | I(1)                 | ambiguous          | 0              | 232            | 7            | 0                        | 7                          | 0      | Low pair overlap in main ... |
| other             | silpo_novus | observed         | daily     | ambiguous            | I(1)                 | I(1)               | 0              | 232            | 56           | 0                        | 52                         | 0      | Low pair overlap in main ... |
| other             | silpo_novus | promo_controlled | daily     | ambiguous            | I(1)                 | I(1)               | 0              | 232            | 56           | 0                        | 52                         | 0      | Low pair overlap in main ... |
| sour_cream        | silpo       | observed         | daily     | I(1)                 | I(1)                 | I(0)               | 1793           | 188            | 48           | 178                      | 38                         | 0      | Admissible for ARDL/ECM s... |
| sour_cream        | silpo       | promo_controlled | daily     | I(1)                 | I(1)                 | I(0)               | 1793           | 188            | 48           | 178                      | 38                         | 0      | Admissible for ARDL/ECM s... |
| sour_cream        | novus       | observed         | daily     | I(1)                 | I(1)                 | ambiguous          | 1793           | 188            | 13           | 178                      | 12                         | 0      | Low pair overlap in main ... |
| sour_cream        | novus       | promo_controlled | daily     | I(1)                 | I(1)                 | ambiguous          | 1793           | 188            | 13           | 178                      | 12                         | 0      | Low pair overlap in main ... |